Comparison with the replicated analysis – Pitz 2025

Details

Experiment design

The study utilized an alpha-design with incomplete blocks to evaluate the transcriptomic landscape of maize heterosis. The experiment comprised three complete biological replicates, within which genotypes were distributed across multiple batches (incomplete blocks) over time to accommodate the large sample size (~1,152 samples). The genetic material included 112 Recombinant Inbred Lines (RILs) and their corresponding testcross hybrids (RIL × B73 and RIL × Mo17), representing varying degrees of heterozygosity. To effectively correct for batch effects and spatial variation within the aeroponic growth systems, the parental inbred lines (B73 and Mo17) were embedded as common checks across all batches and systems.

Assessment of technical variation and family selection

Prior to method validation, we assessed the magnitude of technical variation using the parental inbred lines (B73 and Mo17), which were highly replicated across the experiment. we performed a Variance Component Analysis on the highly replicated parental lines, B73 and Mo17. Using a linear mixed-effects model (~ (1|Replicate) + (1|Batch) + (1|System)), we partitioned the total expression variance into biological and technical components. The analysis revealed that **System** effects were a major source of technical noise, explaining a median of **41.1%** of the variance in B73 and **13.7%** in Mo17. **Batch** effects accounted for an additional **6.4–9.0%** of the variance. In contrast, biological **Replicate** effects contributed minimally (< 2%). These results demonstrate that batch and system technical factors introduce substantial variability. These cannot be ignored in the single replicate analysis.

To validate our single-replicate method without confounding technical variation, we restricted our analysis to families where the hybrid and both parents were processed in the same experimental batch and grown in the exact same aeroponic system. This strict filtering eliminates temporal and spatial noise, ensuring MPH calculations reflect biological differences. We identified a total of 169 such families available for high-precision validation (39 in Replicate 1, 65 in Replicate 2, and 65 in Replicate 3)

Method using single replicate

For each independent replicate run, we treated the data as if it were the only available samples and subset the raw count matrix to include only the samples corresponding to the identified spatially matched families (Hybrid, Maternal Parent, Paternal Parent). A pseudo-count of 1 was added to all raw counts to ensure numerical stability in downstream algorithms. We then performed standard preprocessing using the `edgeR` package: raw counts were normalized using the TMM method to account for sequencing depth, and lowly expressed genes were filtered using the `filterByExpr` function based on Genotype grouping.

We applied our proposed method 2sLRT to the full families in this replicate. For the dispersion estimation stage, parameters were optimized per replicate: for Replicate 1 ($N_0 = 15, \alpha_0 = 0.1, K = 1$); for Replicate 2 ($N_0 = 20, \alpha_0 = 0.01, K = 1$); and for Replicate 3 ($N_0 = 12, \alpha_0 = 0.01, K = 1$), where N_0denotes the minimum number of null families required for inclusion, $\alpha_0$the significance threshold for the Poisson null-family test, and $K$the number of gene clusters. We then compared the MPH calls from each single-replicate run against the results from using the replicates. P-values were adjusted using the Benjamini-Hochberg (FDR) procedure across all genes in each family.

Method using replicates

To establish a "ground truth" for heterosis, we analyzed the full replicated dataset using a rigorous Generalized Linear Mixed Model (GLMM) approach. For fair comparison and computational efficiency, when comparing the results from 2sLRT applied to each replicate, this GLMM approach is applied only to the full families in this replicate.

1. **Preprocessing:** Raw counts were normalized using the TMM method. Genes with low expression were filtered using a design matrix accounting for Genotype to ensure detectability.
2. **Dispersion estimation:** Robust tagwise dispersion parameters (ϕ) were estimated using `edgeR` with the design `~ Genotype + Batch`, utilizing all available replicates to characterize biological noise.
3. **Model specification:** We implemented a GLMM using `glmmTMB` with a Negative Binomial distribution (`nbinom2`). To test for Arithmetic MPH (additive difference) rather than geometric MPH (log-fold change), we utilized a log-link function but performed hypothesis testing on the response scale. The model included fixed effects for Genotype and Batch, and a random intercept for System (`1|System`). Crucially, the dispersion parameter for each gene was fixed to the robust `edgeR` estimate to prevent convergence failures due to the limited sample size of specific genotypes.
4. **Hypothesis testing:** Marginal means were estimated and back-transformed to the arithmetic scale using the `emmeans` package. Significance was determined for the contrast $\mu_{Hybrid} - 0.5(\mu_{P1} + \mu_{P2}) = 0$ using the Delta Method to approximate standard errors. P-values were adjusted using the Benjamini-Hochberg (FDR) procedure across all genes in each family.

Results

Performance Evaluation Framework

To rigorously evaluate the reliability of the 2sLRT single-replicate method, we stratified our analysis based on two dimensions: feature reliability and sample quality. First, we distinguished

between the full transcriptome set, which includes all expressed genes regardless of estimation method, and the high-confidence feature set, a subset comprising genes where dispersion parameters were directly estimated from "null families" rather than predicted or extrapolated by the 2sLRT method. The comparison is at nominal significance level 0.1.

While the method demonstrated acceptable performance on the full transcriptome (Baseline AUC 0.70–0.80), we observed a consistent improvement in discrimination power and ranking accuracy when restricting analysis to the high-confidence set (Baseline AUC 0.72–0.82). Consequently, we report detailed validation results focusing on the high-confidence set to demonstrate the method's optimal capability, though broad application to the full transcriptome remains feasible with slightly reduced precision.

Second, within this high-confidence feature set, we defined three evaluation scenarios to distinguish algorithmic accuracy from data quality limitations:

1. Baseline scenario: The raw output of the method without post-hoc filtering.
2. Practical screening scenario: A subset representing realistic candidate discovery, filtering for quantifiable expression (minimum count ≥ 10 in the hybrid and both parents) and strong signal (heterosis magnitude $> 50\%$ meaning hybrid count is greater than 1.5 mid-parent or smaller than 0.5 of the mid-parent).
3. Theoretical validation scenario: A subset used to validate the statistical model by excluding signal outliers, defined as cases where the single-replicate effect size was inconsistent with the benchmark magnitude (exaggerated by > 2-fold or diluted by $> 50\%$) due to stochastic sampling variance.

1. Global discrimination On the raw dataset, the method demonstrated consistent discriminatory power across all replicates. Replicates 1 and 3 achieved strong global discrimination (AUC **0.82** and **0.80**, respectively). Replicate 2 showed slightly lower baseline performance (AUC **0.72**), reflecting higher stochastic noise in that specific subsample.

2. Practical screening performance In the recommended screening scenario (quantifiable expression, heterosis $> 50\%$), the method functioned as an effective high-pass filter across all sampling conditions. Replicates 1 to 3 achieving **sensitivities** of **74%** and **40.9%, 51.6%** respectively, with **AUC** stays like the raw results, ranging from **0.73** (Replicate 2) to **0.85** (Replicate 3). This variation underscores the expected impact of sampling noise in N=1 designs, yet confirms the method consistently recovers a substantial proportion (40–80%) of strong biological signals for downstream validation.

3. Algorithmic validity To isolate the performance of the statistical model from sampling artifacts, we evaluated the "theoretical validation" subset (excluding stochastic outliers). Remarkably, **all three replicates converged to high performance** on this validated set, with AUC **0.87 for Replicate 1, 0.82 for Replicate 2 and 3**. The fact that Replicate 2 (the noisiest raw dataset) recovered to an AUC of 0.82 when outliers were removed is a critical finding. It confirms that the 2sLRT algorithm is mathematically sound and robust; performance drops in raw data are driven by sampling events (outliers), not by methodological failure.

High confidence genes	Replicate 1			Replicate 2			Replicate 3		
Scenario	AUC	Prec.	Rec.	AUC	Prec.	Rec.	AUC	Prec.	Rec.
Baseline	0.82	21.7%	28.4%	0.72	23.3%	14.3%	0.80	56.6%	22.6%
Practical screening	0.79	26.3%	74%	0.73	20.4%	40.9%	0.85	54.3%	51.6%
Theoretical valid.	0.87	44%	18.7%	0.82	60.1%	13.7%	0.82	87.8%	14%

All genes	Replicate 1			Replicate 2			Replicate 3		
Scenario	AUC	Prec.	Rec.	AUC	Prec.	Rec.	AUC	Prec.	Rec.
Baseline	0.80	21.7%	63.9%	0.70	16.7%	28.3%	0.78	43.9%	28%
Practical screening	0.76	29.1%	95.4%	0.68	14.3%	73.6%	0.75	40.8%	62.1%
Theoretical valid.	0.85	32.6%	67.3%	0.80	33.6%	28.5%	0.79	63.7%	18.4%

Comparison with the replicated analysis – Simulation

We agree that biological replication is the gold standard for RNA-seq analysis and that the lack of replicates poses significant statistical challenges. The reviewer asks a critical question: *Why did we not reduce the number of genotypes to allow for proper biological replication?*

This was a deliberate strategic choice driven by the goals of the study (population-level mapping) rather than purely financial constraints. To validate this choice and address the reviewer's concern about the statistical validity of analyzing unreplicated data, we performed a rigorous new simulation study comparing our proposed method (2sLRT) against a standard replicated benchmark. Details can be found in Qi et al. (2025).

1. Statistical Validity (FDR Control): The reviewer expresses concern that without replicates, it is impossible to distinguish biological signal from noise, potentially leading to false discoveries. Our simulations demonstrate that this is not the case for the 2sLRT.

- We simulated data with $N = 3$ replicates and compared the 2sLRT (using only 1 replicate) against a standard Newton-Raphson Likelihood Ratio Test (using all 3 replicates).
- **Result:** Across all scenarios (including high-dispersion settings derived from plant datasets), the 2sLRT maintained strict False Discovery Rate (FDR) control, consistently keeping the error rate below the nominal 0.05 threshold (observed FDR range: 0.013 – 0.037).
- **Implication:** This confirms that the 2sLRT is statistically valid. It is conservative and does not incorrectly identify random noise as heterosis genes, directly addressing the reviewer's concern about validity.

2. The Strategic Trade-off (Power vs. Sample Size): Regarding the choice to sequence 800 genotypes (H) without replication rather than, for example, 260 genotypes with $N = 3$ replication:

- Our simulations show that while a replicated design achieves near-perfect power ($>$ 99%) for detecting heterosis in a specific family, the 2sLRT achieves respectable sensitivity (60–70%) when the number of families is large ($H = 200$).
- Crucially, we found that the power of the 2sLRT improves as the number of families increases (e.g., TPR rose from ~50% at $H = 100$ to ~70% at $H = 200$).
- **Conclusion:** By including more families rather than replicates, we lose the ability to detect weak effects in specific families, but we gain the ability to map heterosis trends across a much wider genetic diversity.

Summary: Our simulations prove that the statistical testing of heterosis using 2sLRT is robust. The method acts as a high-precision filter: it may miss subtle signals (lower power), but the signals it does detect are statistically reliable (low FDR). This justifies the large-scale unreplicated design for the purpose of identifying high-confidence candidates for population-wide analysis

ngenes	nfamilies	Disp Var	FDR			TPR				pAUC	
			2sLRT	2 Reps	3 Reps	2sLRT	2 Reps	3 Reps	2sLRT	2 Reps	3 Reps
5000	100	low	0.0260	0.0469	0.0460	0.4027	0.9632	0.9965	0.9510	0.9975	0.9998
		high	0.0138	0.0501	0.0481	0.5280	0.9991	0.9997	0.9778	0.9999	1
	150	low	0.0299	0.0477	0.0454	0.5349	0.9654	0.9968	0.9594	0.9977	0.9998
		high	0.0183	0.0497	0.0482	0.6245	0.9994	1.0000	0.9833	1	1
	200	low	0.0370	0.0465	0.0455	0.5948	0.9614	0.9966	0.9621	0.9976	0.9998
		high	0.0214	0.0497	0.0486	0.6842	0.9988	0.9999	0.9854	0.9999	1
10000	100	low	0.0258	0.0471	0.0452	0.3985	0.9631	0.9965	0.9506	0.9976	0.9998
		high	0.0133	0.0487	0.0480	0.5098	0.9990	0.9999	0.9776	0.9999	1
	150	low	0.0284	0.0467	0.0457	0.5344	0.9624	0.9966	0.9596	0.9975	0.9998
		high	0.0182	0.0499	0.0478	0.6268	0.9994	0.9999	0.9831	1	1
	200	low	0.0366	0.0470	0.0453	0.5901	0.9597	0.9969	0.9614	0.9973	0.9998
		high	0.0209	0.0494	0.0477	0.6947	0.9992	0.9999	0.9862	0.9999	1

Comparison with the replicated analysis – In house

To address the reviewer's concern regarding the statistical validity of our unreplicated analysis, we analyzed an in-house dataset comprising 48 samples—representing 10 hybrids and their parental genotypes with two biological replicates—using the negative binomial (NB) framework described by Qi et al. (2025). Data preprocessing included DESeq normalization and the removal of genes containing zero counts in any sample, resulting in a final set of 24,241 genes. We first applied the replicated NB method to the full dataset to establish a benchmark.

To evaluate the performance of the 2sLRT method, we employed a random subsampling strategy repeated 10 times. In each iteration, the dataset was partitioned into two independent sets, ensuring that each set contained exactly one replicate per genotype. We then applied the 2sLRT method to these individual replicates. For dispersion estimation, we used the parameters $N_0 = 2$, $\alpha_0 = 0.01$, and $K = 3$(where $N_0$is the minimum null families, $\alpha_0$is the Poisson null-family test threshold, and $K$is the number of gene clusters).

Using a significance threshold of 0.05 (FDR), we evaluated the performance against the replicated benchmark. The method demonstrated strong concordance, achieving a median AUC of 0.91 (mean = 0.88). Furthermore, the partial AUC (restricted to FPR < 0.2) averaged 0.80, reflecting high sensitivity specifically within the low false-positive range. Although performance varied across random splits (AUC range: 0.64–0.95), the results generally indicate robust predictive power.